

Reihaneh Malekian

Ph.D. Candidate in Statistics, Columbia University
rm3942@columbia.edu | New York, NY

Education

Columbia University, New York, NY

Expected May 2027

Ph.D. in Statistics

Advisor: Sumit Mukherjee

Sharif University of Technology, Tehran, Iran

M.Sc. in Pure Mathematics

Sharif University of Technology, Tehran, Iran

B.Sc. in Mathematics and Applications

Research Interests

Probability and high-dimensional statistics; large deviations; random graphs and hypergraphs; Gibbs measures and interacting systems; empirical Bayes; statistical inference for dependent data.

Publications and Preprints

- **Reihaneh Malekian**, Hao Xing, and Arian Maleki. “Is Speckle Noise More Challenging to Mitigate Than Additive Noise?” *IEEE Transactions on Information Theory*, 71(8):6353–6372, 2025.
- **Reihaneh Malekian** and Aaditya Ramdas. “Matrix Concentration: Order versus Anti-order.” Preprint, arXiv:2408.05998.

Manuscripts in Preparation

- “LDP (Large Deviation Principles) for Tensor Forms,” with Sohom Bhattacharya, Nabarun Deb, and Sumit Mukherjee.
- “Fluctuations in Mean-Field Potts Models,” with Sumit Mukherjee.
- “Statistical Inference for Online Least-Squares SGD under Markov Dependence,” with Promit Ghosal

Selected Research Presentations

- Invited talk, Probability Seminar, UC Berkeley, September 2026.
- Contributed talk, Joint Statistical Meetings, Boston, August 2026.
- Talk, ICERM Hot Topics Workshop, Brown University, August 2026.
- Poster, International Congress of Mathematicians, Philadelphia, July 2026.
- Poster, Peter Hall and Bob Shumway Memorial Conference on Dependent Data Analysis in the Modern Era, UC Davis, May 2026.
- Invited talk, Finger Lakes Probability Seminar, Syracuse University, April 2026.
- Seminar talk, Columbia University Student Seminar, February 2026.
- Talk, Progress in Discrete and Continuous Probability Conference, Union College, Schenectady, NY, July 2025.
- Poster, INFORMS Applied Probability Society Conference, Atlanta, July 2025.
- Poster, Bernoulli–IMS World Congress in Probability and Statistics, Bochum, Germany, August 2024.
- Talk, Minghui Yu Memorial Conference, Columbia University, April 2024, April 2025 and April 2026.

Honors and Awards

- Teaching Observation Fellow, Columbia University Center for Teaching and Learning, 2026–2027.
- Early Career Travel Award, American Statistical Association, 2026.
- Graduate Research Fellowship, Columbia University, 2022–2027.
- Ranked second in overall GPA in both the B.Sc. and M.Sc. cohorts, Sharif University of Technology.
- Recipient of undergraduate and graduate scholarships from Iran’s National Elites Foundation.
- Admitted to the graduate program without an entrance examination as an exceptionally talented B.Sc. student, Sharif University of Technology.
- Placed among the top 0.05% of participants in Iran’s nationwide university entrance examination.

Selected Workshops and Summer Schools

ICERM Hot Topics Workshop

Synergies between Geometry, Probability, and Computation in High Dimensions, Brown University August 2026

SLMath Summer Graduate School

Concentration Inequalities and Localization Techniques in High-Dimensional Probability and Geometry July 2023

Teaching Experience

Teaching Assistant, Columbia University 2022–Present

- Introduction to Statistics; Statistical Inference; Linear Regression Models; Introduction to Statistical Reasoning; Introduction to Probability and Statistics; Probability.

Teaching Assistant, Sharif University of Technology

- Probability and Applications; Probability Theory; Mathematical Writing; Algebra I; Analysis I.

Research Mentoring

Bachelor’s Project Co-supervisor, EPFL 2021

Co-supervised Clémence Marcaillou’s bachelor’s project in combinatorial discrepancy theory through weekly research meetings.

Selected Expository Writing

- **Reihaneh Malekian**. “Good Numbers.” *Pargar*, Year 2, no. 4, pp. 17–23, Spring 2014. [In Persian]
- **Reihaneh Malekian** (translator and adapter). “The Turán Number and Probabilistic Combinatorics,” by Alan J. Aw. Persian translation and adaptation of the article published in *The American Mathematical Monthly*, 119(6):510–513, 2012. *Pargar*, Year 2, no. 8, pp. 9–13, Spring 2015. [In Persian]

Outreach and Academic Service

Organizing Team Member and Program Moderator, Maryam Mirzakhani Celebration May 2019

Sharif University of Technology, Tehran, Iran

Student Organizer, Undergraduate Mathematics Summer School, IASBS, Zanjan, Iran 2019

Student Organizer, Mathematics Summer School for High School Students, Borujen, Iran 2015

Skills

Programming: Python, R.

Languages: Persian (Farsi; native), English (proficient), French (intermediate, B1).